

# ActInf GuestStream 065.1 ~ Phoebe Klett, "Towards Bayesian World Models"

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hello and welcome this is active inference guest stream 65.1 it's December 6th 2023 we'll be hearing from Phoebe Klett and Dan Simpson on Bayesian world models for explainable transparent reasoning there will be a presentation followed by a discussion so thank you for joining Phoebe very much looking forward to the presentation to you awesome thanks for having us uh yeah and I'm really excited to chat with you all about how we might start to integrate today's state-of-the-art language models um into more probabilistic machinery and what that might buy us all right let's get right into it um all right so some of the things that I'm hoping to discuss today uh include why we might use a language model for something that isn't long form text generation uh and how we might do that uh and then motivate a little bit kind of why we might need a world model um what simple uh self-organizing World models might look like uh and even in the simplest cases how we might start to use those um as effective recommendation engines in the wild today uh and then a little bit of discussion about kind of where this research is going all right so what are language models good at uh today's uh especially large language models um are trained on next token production uh so this means given a sequence of tokens uh we're going to estimate um which token is most likely to come next uh maybe rephrasing that a little bit um we might also say that language models uh are trained to estimate uh which sequences of tokens or words are likely uh the caveat being likely to appear in the training set um but today our training sets are quite extensive so given this simple training objective it's um our arguably surprising uh that we've seen language models do as well as they have uh in in really impressive tasks uh so they start to demonstrate like really great um language understanding meaning like the semantics of language itself uh not just the syntax of kind of like which sentences make sense um they've also started to demonstrate some uh World Knowledge so implicitly learning kind of how the world works just through our own human abstraction um and how we uh articulate that uh we start to see language models falter at tasks um which look more like symbolic problem solving uh so math

in particular we see this in programming although we're getting better at this uh and in general any kind of like long-term planning tasks which require abstract reasoning um so we see this also in problems that look like this which are word problems um but which are really about kind of like understanding abstractly how these kind of abstract objects relate to each other uh and again this shouldn't be surprising that language models struggle to do to do things like this um since it's so far off from their original training objective um and in particular even when we arrive at the right answer in some of these cases it's really hard to know after the fact kind of the reasoning for how we ended up at the right answer in this particular case uh and maybe more fundamentally uh if we don't know where the abstraction is happening or where the reasoning is happening it's very hard to guide that process uh and so this starts to motivate the need for something that looks more like an explicit World model uh and now to start to borrow some uh words from Yasha Benjo this is one of the um kind of biggest issues with today's language models is arguably that we're asking the language model to be both the inference machine and the world model implicitly uh when this uh doesn't quite make sense um and so some things that we might hope for in a world model uh are to model causal relationships to be really adapt at modeling using uncertainty uh and to be a modular um Yan Lun puts this in a in a similar way so uh we might kind of ask a world model uh to be able to distinguish between which details are important versus irrelevant uh and to be able to make predictions uh that can be performed in sort of like this abstract space of representations um and so hopefully in the next uh following slides we're going to motivate um what it might look like to use language models as the inference machines in maybe like the simplest case of self-organizing world models um and even in those most simple cases how we start to get uh something much closer to the kind of uh explainable reasoning um which I think lots of us are grasping for right now all right so in a very simple case we might think of a world model as simply a collection of hypotheses where we model confidence um over over each of those hypotheses uh and in particular we care about predictive World models so given some evidence or data that we've observed in the world we'd like to propose uh the the world models um which best explain uh the evidence that we've observed uh and if that language s sounded leading uh it was indeed we are proposing in this case to use a Bayes rule which we all know and love um which tells us exactly how to update our beliefs given some new evidence uh and so uh when our hypotheses uh are kind of these uh purely random variables this bottom term uh can simply be expanded um into this this equation on the slide and so the the real tricky part in this computation uh is the likelihood piece um so given this evidence given that uh our hypothesis is true what's the

likelihood that we would have observed the evidence uh and the the claim that we're making is that language models are actually this is this is like a natural task task of a language model um when our evidence and our hypothesis are both semantic objects as we discussed language models are kind of trained exactly to understand which sequences of text are likely um so asking it to prompting it in a clever way such that we can extract uh this particular likelihood is actually really natural so how do we do this um so we come up with this kind of clever prompting scheme um that allows us to extract again exactly How likely uh is is this to have C so given that our hypothesis is H in this setting uh and our evidence is this uh cly e um we only need to change our evidence into the conditional form and then phrase the question like this to the to the language model so the input sequence of text uh goes is like given that you know potentially if our hypothesis is like Walmart has been sely impacted by covid-19 pandemic would we have observed the evidence um that Walmart laid off 10% of the material staff um and so either using something like f shot prompting or guidance or some kind of control generation technique um we can ensure that our language model outputs either yes or no and then uh use the logits from its answer uh to at that probability um and again the the claim here is that like this is actually really natural task um to ask of a language model um that leverages uh its like innate reasoning engine uh better than just kind of allowing it to ramble uh using using text sometimes and we can even make this updating scheme um a bit more sophisticated using things like Precision waiting um so this will buy our our posterior over hypothesis um either towards our prior or our evidence that we've observed um based on how confident we are in our prior and in our evidence all right so the world models that we've seen so far um are fairly simple uh in particular we're modeling all our hypotheses as independent from each other uh and this seems like a a large simplifying assumption um so how might we support more complex World models where we condition our hypotheses on each other all right so another thing that um hopefully will be familiar to this crowd basinets have been around for forever and um are often used to model mechanistic failures and en llarge systems um and can be understood simply as distribution or each variable depends on some small number of ancestor variables uh perhaps more intuitively we you can also think these think of these as directed graphs um where we have edges between variables um which are directed in the sense that the parent is conditioned on the child so let's see an example all right so now we have uh two different kinds of hypotheses uh the ones which are kind of parents in our simple setup um are kind of qualitatively more abstract than the children um so our two children nudes um are similar to the hypothesis we saw before um Market sentiment for Best Buy poor

or Walmart will grow its physical footprint this year uh and then the more abstract hypothesis being retailers were negatively infected by covid-19 um and so then we specify uh this conditional structure um either kind of in a classical sense by specifying all of the joint probabilities up front um or alternatively learning it given some healthy amount of train data uh and again we're proposing that instead of doing that uh intensive process we can use language models to extract those probabilities in a natural way um and so one natural complaint at this point might be like well this space is going to start to get very large um if we're trying to update these beliefs in real time given some large data stream this is going to become intractable uh given our current framework um so how might we start to augment that luckily there's been a lot of work historically done on this problem uh and we can start to use things like message passing to update our beliefs in an approximate way um so as long as our basen has a tree dependency structure we can use things like the Sum-Product uh algorithm to update our beliefs and I don't want you to worry too much about the equations on this slide if you've seen this before uh this kind of recursive structure will look familiar but if not the general kind of intuitive idea that we're going to compute these messages from all of the children or the neighboring nodes and use those to propagate through the graph to update um each independent belief but let's just see an example all right so suppose we observe some new evidence and we'd like to know how should we update the probability um over hypothesis  $a$  which as you remember is the parent node in our graph um all right so  $B$  here on the left hand side is the belief um  $X_A$  is essentially like just the very able representing the hypothesis  $a$  um and then the  $F_e$  and the side terms are specified um by our language model or the um the Basen in the classical sense uh and so we're summing over all of the variables all of the values that the variables  $X_B$  and  $X_C$  can take which in our case of course is just zero and one um so if you uh you might recognize at this point that this is indeed the exact marginal probability um for the hypothesis  $a$  um and that's because our graph is Simply Connected in this example so in general this isn't true but um turns out to be the case in in our example and again uh we can compute these  $F_e$  and  $S$  terms uh using the language model itself all right so because we're all levers of free energy here going to walk through kind of how this is uh maybe the first example of uh a self-evidencing um or minimizing free energy um kind of model so um as we noted belief propagation isn't exact for more complicated graphs and so uh it makes sense and might be useful to ask the question um you know how far apart or when are our beliefs um close to the exact marginals uh and so we often use um things like Kullback Divergence to compare the difference between two probability distributions and that um is explicated uh on the right

hand side uh and then those of us um who who are uh like have a background in physics might recognize boltzman's law as well um so this is just the idea that we might represent the probability of a given State using an energy function uh and we're not going to accept this as truth maybe as some of us have done in the past um but we are going to just use it as a definition for this energy function such that when we plug in that term here um and expand out we start to see these first two terms like a lot like the kind of energy and entropy functions which we are used to and indeed we can just uh classify those two terms uh as the Gibbs free energy function um yeah which makes me happy to to see this all coming together uh and in particular it might make sense uh just to note at this point that using um world models which are self-organizing in this sense uh seems to be very compelling since the kind of world model which we want is one um which promotes the evidence that we've seen so far um why is it uh useful to to formulate this in terms of free energy besides the fact that we all um find it compelling here uh well you can make a lot of progress by constructing analytically tractable approximations of Gibbs free energy often um I'm not going to go into the details here uh but here are two examples where for that been fruitful all right great um so now I'm going to chat briefly about kind of how we might use recommendation systems like or how we might use these systems as recommendation systems uh and indeed in the world uh today all right so one like prime example for aware system like this might be useful is is a situation where we have kind of lots of data incoming uh at very high frequencies and we always want to have some set of like naturally discrete hypotheses that we're modeling beliefs over um which are being kept up to dat um at a at a very regular Cadence uh and so actually a lot of the the muscle here is just reformatting documents or however our data comes in as evidences this is not always obvious or easy to do um but once you've kind of figured out that part uh and in particular um we've been using things like rag retrieval augmented generation or um embedding based systems to kind of figure out when data that's in in is relevant to a given hypothesis um once you've kind of built up that Machinery the actual updating computations as we've shown um already is actually pretty simple so we do these likelihood computations and we update our beliefs uh and then at any given time we can query that model um for our marginal distribution um over any given hypothesis uh and it turns out that this um kind of setup has many practical applications it's also noteworthy that even with very simple systems like these um these are like out of the box controllable and explainable um so just by storing the magnitude and the direction of the update to the posterior for each piece of evidence that we observe we have a very natural built-in explanation for our belief at any given time uh and that makes kind of like

these applications where folks might really like to use a language model but really require like a robust like causal relationship between the outputs and the explanation um which you don't get from a language model on its own a system like this can be very appealing in those situations all right so now on to further work so everything that we've discussed today um is early work towards um integrating language models uh into more probabilistic Frameworks uh and there's been a lot of exciting work done uh in this vein right now um some important questions which are especially interesting to me or which parts of the world model should be learned versus encoded um and how do we want intelligence to scale so I mean that uh both in the sense of composing systems naturally there should be some very like natural way that we can compose to intelligent systems and also such that we can scale them with compute uh and I don't mean to restrict myself either to the kinds of compute that we have today we're also working at some uh exciting new Computing paradigms at normal um which might be more compatible with software uh of this nature um also the two folks that we referenced at the beginning of the talk Jeff Hinton and Yan Lun have done really exciting work in this area um which is yeah very inspiring and so in particular G Nets are also probabilistic graphical models um which I think folks will find um a natural next step in Reading uh yeah if if you so desire and that's um that's it for me awesome thank you wow very cool Dan do you want to give a first reflection or thought and then meanwhile anyone who's watching live please feel free to write questions I'll relay them in um absolutely um so hi I'm I'm I'm Dan I work with Phoebe on this project um and yes the um I think the thing that's most exciting about this for me personally is sort of twofold one of them is that it's a way of avoiding sort of having to trust a language model to understand and reason about text um because they're not it's not that they're bad the thing is the the extremely strange thing about language models is they're quite good at being almost good enough um but they're never quite what you could use you could never use a language model to I don't know uh sort of triage like an important sort of situation where a bunch of different things are coming in you have to make a decision about which is important the reason you can't do that is you you simply cannot understand the encoded biases you cannot get it to reliably generate reasoning you can ask it for reasoning but the thing that it prints out is not the reasoning that it used internally because it doesn't reason um fundamentally while these have input and output that are natural language they are not artificially intelligent they are just prediction machines um and so we have to be very careful about not anthropomorphizing them so this is a way of using those incredibly powerful prediction machines in a framework where we can um make sure that we uh essentially keep a

record of what we're doing so the human can look at it because I mean there's a lot of talk in this world about sort of posthuman AI and those sorts of things the idea that the machines will become intelligent enough or the machines will rise up in a slightly more alarming Type of Way um and that's all great and wonderful but that's not particularly interesting to us at normal we're much more interested in sort of having you know mimicking explicit decision processes so that a human can audit them and can make these things work so that's kind of the like the area that we're we're sort of coming from awesome all right I'll go to a question from the chat so Josh asks great talk where does hypothesis relevance enter the calculus is it folded into confidence not sure if it ought to be just saw hypothesis relevance mentioned hypothesis relevance does that mean um like which hypotheses are conditioned on each other is it possible to ask a clarifying question there maybe Dan you have a they they can follow up but yeah like um I also wondered about this like you might know what was relevant maybe the temperature and the rainfall were relevant but then how does this approach help us understand when one of those relevant factors no longer is relevant or when a new relevant Factor comes into play yeah these are great questions um so I think in terms of understanding in an automatic sense when two hypotheses are are relevant to each other we can leverage um embedding type you know language models for this kind of thing also um if we don't have a more kind of like structured human intuitive sense for when to hypotheses are are related uh in terms of like how those relationships evolve over time this is something that's really interesting to me and I think looking at the the theory behind structure learning or when we propose to add new nodes to the network or propos to add a new Edge to the network or things like this uh is a really exciting research Direction um although I I don't have like a silver bullet answer to to how we should do that um just to like add a little bit more to that um it is like it is a really interesting research Direction um like one of one of the things that Phoebe mentioned in the talk is that there is a difficult step that we're not talking about which is actually translating this natural language into reasonable hypotheses so there is a step in there where you take essentially a chunk of text and you have to decide if this is a hypothesis if this is a hypothesis we've seen before if this is a sub hypothesis or a clarification of a hypothesis that we've seen before and so on and so forth so that's in some sense part of the data processing and it's it is an important step and one that we are sort of continuing to work on and refine uh the other thing like a different sort of interpretation of the question around relevance is um around sort of is the hypothesis relevant to the thing that you're looking at I mean we could have a hypothesis that the sky is blue but if we are deciding to deciding you know whether whether or

not we need to check that car's oil like the the the truth or not of the color of the sky is very irrelevant and that then comes into the nice thing about having your world described as a um collection of statements with truth values associated with them in that you can directly reason over them so you can put a classical decision framework um over that to take into account both the sort of the knowledge you have of the world and also like which parts of these worlds are are sort of unknown so in that sort of situation the person using the world model to Constructor um a a sort of decision or an output will be responsible in some sense for assigning a weight or a cost to each hypothesis being true and for some of those hypotheses obviously it will be zero um because again we do not care about the color of the sky if all I want to know is if I need to change the oil in my car um so that's the that's the sort of the other end of the answer so there's a version of the answer at the start of the information flow and there's a version of the answer at the end of the information flow but it is a tricky uh point and one that we are sort of continuing to iterate on to try and find sort of good ways on both ends of that yeah well a lot there um it's very interesting how in that presentation in response I heard both about probability distributions on rules and rules on probability distributions and like with which one whether it's the tail wagging the dog or the horse in the cart how to design these synthetic intelligence systems that appropriately bring together aspects that are more symbolic more rule like and then more probabilistic more embedding like so where does that land with you or how do you see the design of these systems with mixed symbolic and probabilistic components yeah yeah that's a great point and I think this really gets it like which parts of the world model should be learned um or should be represented in some like more discrete space versus like encoded based on our own human intuition for rules and structure um and I think like maybe this is would be fairly represented as a copout answer but I think it depends a lot on the application uh I think like when we're developing systems like this and and just trying to iterate through as many different um hypotheses uh as you can quickly um like choosing an application and and benchmarking and testing and seeing like what actually works um is is a go-to strategy for us uh in terms of like well which parts are should be fixed and are actually helpful um in to increase reliability such that like we can use our human intuition for how this particular you know system is built uh versus like well this is something that we we want uncertainty over that's like a really important part of the learning process for us in terms of um yeah that kind of iteration um so I think I think it probably depends on the application yeah it's um it it definitely depends on the application it's also um like depends on where the actual challenge points are so we've got like outside of this we've got sort of a few



other things that we've released publicly that kind of look at this idea of there being like external rules to the uh um to the the system and whether or not we can add those in so one of them is something called constrained generation where we we sort of force the model to only produce something valid and that's sort of quite a useful way of removing one particular aspect of stress to from the model which is that it may make sort of syntactically or sort of in somehow incoherent um outputs that don't follow the rules and then we can then focus with the rest of our energy we can then take that as given and focus with the rest of our energy on improving the bit that we don't have rules for um so so those sorts of things and sort of a different version is trying to improve something by saying no you broke a rule we need to like go back and make this not sort of true so this kind of um of Chain of Thought prompting type of idea um so so yeah the um the symbolic and probabilistic um I think in our minds live very closely together as two tools that don't completely solve the same problem um and I think there's sort of in in the world of I'm not sure how familiar anyone in the audience is with language modeling but like in the world of language modeling before this sort of explosion of neural networks and artificial int type methods uh there was a lot of work on symbolics of language and grammars and all of that sort of stuff and that work pushed quite a long way forward um and this work is pushing quite a long way forward and I suspect the next thing is going to involve them joining up again uh because they each have good points they each have bad points and you know two wrongs don't necessarily make her right but they can make her less wrong nice yeah recently we heard from Elliot Murphy and talking about the neural Linguistics and about how the statistics of language are not the rules of language you can always come up with a new expression that's never been uttered that's not going to be in the training distribution or any distribution MH okay I'll ask a question in the chat from upcycle Club they wrote what are some of the key challenges associated with developing such basian World models hm I think we've touched on a bunch of them um the ones that are most top of mind for me right now are are this the structure learning thing that came up so how do we understand like when to propose new hypotheses and how to integrate those into the models um and then yeah just figuring out like yeah I I guess this like proposal and evolution process of the nodes themselves um since everything else like the framework like works pretty automatically and in a reasonable way uh thank you B thank you to the development of language models um but kind of moving from this like more uh discret case into a continuous case um which like more fully represents the space that we're learning over um can be challenging yeah um I would also say that like it's a sort of a saying let's go in that direction uh

there's a c saying in this world that um that sort of no model ever survives its first encounter with data um and that that becomes true here as well so there's lots of like as we've been building these things and using them we found lots of little spiky Edge Corners with sort of making sure that the language model is actually doing what we wanted to do so there are a lot of questions in building these things around how do you actually test that the the the components of it are actually working the way you want and then on like a broader level how do you compare something that is fundamentally trying to solve a different problem um to other methods so we are solving a problem under the constraints that we want a fully auditable system um we could also solve all of these problems by a thing called in context learning which is basically putting the context into the prompt of a large language model and asking it the answer and that also works uh especially when you've got things like GPT 4 which are just wonders and glories um it works really well so then we come to the question of how do we actually make the case for this from a like a bigger picture perspective can it be more than just a like can we find benchmarks that reflect um the structural advantages of this approach over something like in context learning uh that don't come across as false so that's kind of like a a stranger answer because it's not really about like actually developing the world model it's about sort of convincing other people that it's a good idea um and that's you know that that is a thing that is true of essentially all of the things on this slide as well they are all quite complex and odd little methods um that you know there's a there's a degree to which well we definitely can solve this an easier way um so what is the thing that what is the the the application or The Benchmark where we can say no if you do it the easier way you will fail at this measurable thing very interesting um so you mentioned the self evidencing advantages of using World models that are self- evidencing rather than reward maximizing for example so how do you see that playing out and I can connect it back to active inference of course but how do you see this self- evidencing centrality play out in the kinds of models described here yeah I think there are a couple reasons why it's so compelling to me uh and the first just has to do with explainability right like it's really convincing to people to say like well why why did we predict this why do we believe this well the this is the actual real world data that we've observed such that you know this this is the impact that that's had uh and then I think like uh I don't know you hear a lot about like designing these really complicated reward functions which are often very clever but which um often I feel like are close to being a pitfall because they very easily become like disconnected from like the complex world that we're trying to model and so you end up in like weird local maximums or minimums and um yeah you start

like just kind of uh solving the problem that you've designed versus like the problem which actually exists and so um I I just have always left the idea that what we should be doing is um kind of self-evidencing and from an intuitive sense that feels like what it feels like what an intelligence system should do uh yeah yeah I actually don't have anything interesting to add to that I just agree with fa that explainability and the capacity to explicitly reference previous data including like leave one out so techniques from non-parametric statistics about the effect of adding in another piece of data or removing a piece of data um and then just like to bring it to like a homeostatic setting which is a commonly considered an active inference like we're trying to be within a homeostatic temperature range of 37 yes we could propose reward functions but as those start to include open-endedness and exploration structure learning just like you described it phbe like we're solving the problem as designed rather than the actual question of the homeostatic temperature and the sort of path of least action first principles physics grounded intelligence perspective from active inference is like make it the kind of thing that measures itself at 37 and then as long as it is it is and when it isn't it's dead and that's the kind of mortal Computing crossover which is like outside of its zone of surprise it's not just that it's getting a bad grade in the class that is like a deeper failure signal than that and to understand okay when is it a yellow flag when is it a red flag in terms of the new scientific literature coming in those have plain straightforward ways to interpret that developing larger higher order apparatuses will never turn to that kind of Basil Simplicity yeah couldn't agree more yeah AB I mean absolutely it also like the other thing that it can do quite well is deal essentially outlier studies so SI situations where you have a new piece of information that is strongly conflict with all the previous pieces of information um and trying to sort of work through what that really means um and there's a like there's a degree to which uh we can even sort of extend this process uh to multiple agents that have these belief systems and then look at sort of consensus of experts or weighted consensus of experts so for instance you could have like a we vein type of situation where somebody really over indexes to every new piece information um and you would do that with you know technically you do that with maybe a power posterior type thing or you can have somebody who's built in strong priors in a particular direction um and you can then like take your consensus of artificial um sort of decision making all of which has within their Universe well reasoned updates to the data and then you can look and try and work out what that swarm of experts can tell you um and sort of do very empirical things like try and you know work out which of these experts is doing well at a particular moment in time um because you know there could be there could be times when

the world's very or the problem that you're solving is very chaotic in which case the over indexing uh expert would probably be a pretty pretty solid bet um while there are other times where sort of things are pretty stable and you probably it would be possible that the um sort of the more conservative expert is more uh sort of empirically making good decisions and good recommendations so there's like a lot of ways that we can not just like incorporate these sort homeostasis type ideas but we can also change what that means for different agents and artificers like do that artificially and then combine them together to try and get a almost like a um uh like blanking on the word but you know um a forecast under a sort of a hypothetical set of situations and we can actually sort of bring those ideas of the world forward and see what happens when they sort of meet with actual information yeah this angle of mixture of experts as it's sometimes called more in the language model space or ecosystems of shared intelligence or diverse intelligences in the active inference area like that's very interesting obviously has connections Back To Human teams and teams of Beyond humans and so on um a lot of this is still text based so maybe you did or didn't mention what representation the base graphs are but they're plain text like and there was a lot of discussion about bringing from natural language scientific papers or however it is into a structured form and then the explain method that you showed kind of take the structured form and just giving a little syntactic fluency so it looks human readable so how do you see that Essence coming into play with multimodal models and then with action in the world that isn't just developing the next text token but uh robotic actuator or modifying some other control element of the world yeah that's a really good question um I honestly haven't thought much about multimodal stuff in this particular context but I think um I think the framework is is General enough at this point such that it it's definitely could support lots of different modalities um I I'm yeah I'd be really curious to see how this did with something like audio in particular uh yeah and then um to your point about like uh yeah this maybe like Reet versus continuous um relationship I think I think that's like part of what we're we're learning is how to go from like these long natural text documents to a system which is appropriately discretizing um our hypotheses um such that we we have these like meaningful explanations like you mentioned um so I think yeah I think like continuing to develop like uh robust ways of um surfacing those explanations is a big part of this as well um like over time we're going to observe lots and lots and lots of evidence um how do we make sure like hypotheses don't get stale and how do we use evidence to know when they are and and things like this um are part of that also I don't know if that directly answered your question but that's some of the stuff that I've been thinking about related to that so in the I

I mean in examples like sort of moving towards optics and sort of tech video generation and image generation other audio other sort of multimodalities um to be honest I think of these processes in general as enabling a like building a world model to enable a sort of sequential decision process um so if that decision process happens to be should the robot turn left and that's what the sort of the decision proc processes it's it's multimodal in like a very classical sense that you can put any type of decision framework over the top but it's not sort of generatively multimodal I'm not saying write me a song that sounds like Beyonce and a song that sounds like Beyonce comes out I think this this sort of this sort of basian world model layer is blocking towards that sort of thing but that's that's really not sort of the aim of what we're trying to do it's also like within normal like our almost I I don't want I don't want to say Mantra or Manifesto because that sounds culty and no one wants to sound culty but like our our basic aim is to always Center like humans within our process um and so some of this multimodal stuff it's less clear where the human lives so for instance like a video generation type thing where does the human live so keeping it at this abstraction of sequential decision making then it's a decision that a that a human could do you know human with their thumbs could be moving like a robot around and doing that sort of stuff um but yeah it's it's really all about sort of controllability and auditability for us in sort of a sequential decision process so to the extent that that sort of leads in its multimodal world that that's um that that's sort of part of what we're doing and like to some some versions of multimodality um is we're we're just not swimming in that particular space um yeah not a great answer but a long one let them distill it down later um in the um auditability area it almost falls out to me to be like a synta taxa of auditability and a semantics at the syntactic level just tagging or versioning when a file comes in or when a given computation is executed that is basically transfer across all settings and then where I see you honing in on with with this work is kind of the semantic auditability which is actually how we compose our accounts I would have driven but I decided to walk because this happen and so bringing that different kind of Trace to to systems is going to make it um what will it open up in science or education or how do you see this sitting at a consle somebody is at now and making this different like over what timeline yeah I mean it's really quite nice for storytelling because as you said you can say things precisely like well you know because we observed this thing or because if we had observed something else you know like maybe you can even make statements which are um yeah conditional in that sense uh I I think it does like Empower whoever is sitting in front of this data to feel like really sure about again like the the reasoning and that like that went on uh which to me is is pretty different from what

it feels like to sit in front of chat gpt even though it's quite useful often um you know you you try the code and it works or it doesn't work or you like you know ask your friend is this really true um and that feels pretty different to me from being able to look at the evidences themselves and say like oh well actually if this is the reason you think that I know that that evidence is is not true or you know like you can bring your own human intuition or World model uh in terms of validating or um yeah superimposing what you believe on top of what this system believes and so that makes it really easy to make decisions um quickly I think there's also sort of a Converse of this which is that it also makes it clear which evidence was not used to make a decision uh and that can be quite telling in these situations where you could be worried that a particular type of evidence isn't being weighted correctly or isn't being um sort of formatted correctly so again like if this is a sort of a like a system that builds an assistant um that sort of does surfaces all this information and sort of makes a recommendation with reasoning for a person that person can then look and be like and they know what the data is they can look at the D and say you know why didn't you consider the make of the car or why didn't you consider this or why didn't you consider that and they can then use their understanding of what's not being prominently used by the model to sort of sense test like it's it's sort of I mean in some sense that usage of it is a reformulation of what Phoebe just said where you use your internal World model but it's like I think it's it's important to know when evidence is being used and this is like a thing you simply cannot get um from from like a GPT type thing or any sort of like prompting type method we know for instance that like um the order of the order that you submit your evidence in is probably going to matter for a prompting based method okay that's obviously not true for a basian update where we have this sort of this coherence principle where if you shuffle your data and enter it in a different way you will get the same posterior up to computational artifacts um so so so all of that is like in my mind is just as important to orderability as the ability to write a report that says I made this decision for these reasons yeah well that makes me think about this kind of view from the inside interpretability where the rules help and also knowing what evidence is not used is importance for compliance and knowing what information like in a healthcare setting was or wasn't used um what about thermodynamics we heard about free energy boltzman came up how do you see the info Thermo Nexus what have we learned from the last 100 years of thermodynamic dynamics and information Theory and all of this and on the software or Hardware side how is that kind of a free energy Nexus being used yeah I mean I I'm really excited about how all of this seems to be coming together um I think free energy just

keeps showing up in all of these exciting areas to me we have like a book club for singular learning Theory and like they talk all about free energy too and I think some of those ideas are really exciting um I mean at normal I think like the thing that I would highlight is like this idea of software Hardware codesign um which is really special uh and so we're trying to do this hard and fun dance towards each other where we're like thinking about these new kinds of systems and how they might support each other and um Empower each other and and yeah how to build full stack systems um which is really challenging and and also really exciting um yeah and I think like from like a first principles thermodynamics perspective like like we're all just uh yeah we're all kind of like mathematics and physics people at heart so like going taking like uh you know all of what people have learned in language modeling and all of that like um very much to heart as well like I think approaching whatever problem that we're facing from a first principles how do physical systems work in the world what do we really what are the assumptions we're really comfortable with uh and building up from there um is definitely our our natural mode uh so I think that makes it easier to to start working together also um it's also probably worth saying that we have a sort of a secondary well not secondary a very different stream of interest in thermodynamics as well which is the ways that we can use actual physical thermodynamical principles to build Hardware that is specifically has sort of noise in it as a first class Citizen and because of that it is particularly well suited to tasks um and so we've we've built if you if anyone wants to look we have a Blog I believe the url's blog. normal computing. um and amongst other things that are on it uh there is the very first demonstration of using physical thermodynamic Hardware to actually do computations um is the computation the most vital computation that we will ever do it's inverting an 8 by 8 Matrix so no we can do that other ways um but it is sort of building up towards this idea that we can use thermodynamics not just in our modeling and our understanding of the world but also in our sort of low energy compute stack to actually realize these things um so I think think I it's I think it would be challenging to find a group of people on this Earth who have more investment in thermodynamics and don't work in a physics department uh because we have investment all the way through from sort of active influence type things all the way down to this like this this hot thing goes there which is kind of cool not a physicist so I have my that's my view that's my understanding of thermodynamics this hot thing goes there informative thing goes here hot thing over there call it a day um yep it's a really cool fusion with the kind of parsimony and elegance and the aesthetic of math and physics and first principles and the different parsimony of pragmatism with the actual material basis like of a synapse the size of

the synapse and the kind of stochasticity that that size Alone um entails with like membranes and all this those stochastic aspects are leveraged for the compute the synopsis is not simply a variance reducing machine and so it's like both the platonic SL mathematical is Spirit finds a common home in these real simple physical demonstrations and um today it feels like there's a big gap between the um misos scale computational architectures that you describe today that are very much running on the kind of Von noyman architecture Turing completeness Paradigm and yet very tantalizingly close like to a physical object that has a constrained rule de facto like only one thing can come out of this at a time as long as the funnel is this wide and so bring the rules and the regularities of what we call physical things to bear with the fundamental and the imposed constraints on informational spaces it's very cool directions um one other note about just where active inference um and action plays a role is um and also you mentioned like the hypotheses going stale or like sort of data being over relied on um in in the proactive stance where we're using expected free energy or something like it to to calculate future courses of action over observations that we haven't seen yet moves that haven't been made yet there's an explicit epistemic value and so that can be diagnosed and observed as a measure of where a given computation is on the Continuum between purely pragmatic value just constraint satisficing and realizing preferences and expectations and then the pure epistemic value where all outcomes are good and the more information gain the better and then being able to take control of that balance and know amidst changing situations again taking probabilistic or rule-based approach there to when epistemic and pragmatic like gas and break kind of come into play these are very basil um control knobs or features in active inference that it's just not going to show up at the 50th layer of scaling is all you need yeah yep cool well do you have any other like thoughts or things you want to add or questions or where things are heading for your Works nothing to add at this moment but certainly uh excited to keep in touch with this community and yeah collaborating yeah absolutely um and we sort of share I mean we write papers and stuff but we mostly like we share most of what we do uh be it academic in the sort of machine learning space or be it in the um sort of the physical Hardware space uh on our blog which is blog. normal computing. um and yeah thank you so much for inviting us it's been very fun awesome thank you hope to speak again so peace bye thanks bye